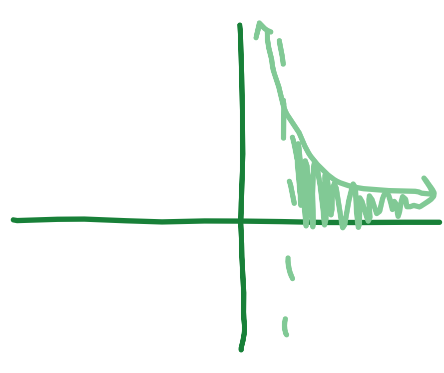
Improper Integrals

section 8.8
2-11-26

A) Area to ∞

Ex:

$$\int_1^\infty \frac{1}{x^2}\,dx =$$



$$\lim_{a\to\infty} -\frac{1}{x}\Big|_1^a = \lim_{a\to\infty}\left(-\frac{1}{a}+1\right) = 1$$

$\boxed{\text{converges to } 1}$

Ex:

$$\int_0^\infty e^{-2x}\,dx =$$

$$\lim_{a\to\infty} -\frac{e^{-2x}}{2}\Big|_0^a = \lim_{a\to\infty}\left(-\frac{e^{-2a}}{2}+\frac{1}{2}\right) = \frac{1}{2}$$

$\boxed{\text{converges to } \frac{1}{2}}$

Ex:

$$\int_{-\infty}^\infty \frac{1}{1+x^2}\,dx =$$

$$= \int_{-\infty}^0 \frac{1}{1+x^2}\,dx + \int_0^\infty \frac{1}{1+x^2}\,dx$$

$$= 2\int_0^\infty \frac{1}{1+x^2}\,dx = \lim_{a\to\infty} 2\arctan(x)\Big|_0^a$$

$$= \lim_{a\to\infty} 2\arctan(a)$$

$$= 2\left(\pi/2\right)$$

$\boxed{\text{converges to } \pi}$

Ex:

$$\int_{-\infty}^0 xe^{-2x}\,dx =$$

	u	dv
+	x	e^{-2x}
−	1	$-\frac{1}{2}e^{-2x}$
+	0	$\frac{1}{4}e^{-2x}$

$$= \lim_{a\to-\infty} -\frac{x}{2}e^{-2x} - \frac{1}{4}e^{-2x}\Big|_a^0$$

$$= \lim_{a\to-\infty}\left(-\frac{1}{4}+\left(\frac{a}{2}e^{-2a}+\frac{1}{4}e^{-2a}\right)\right)$$

$$= \lim_{a\to-\infty}\left(-\frac{1}{4}+e^{-2a}\left(\frac{a}{2}+\frac{1}{4}\right)\right)$$

$\boxed{\text{Diverges to } -\infty}$

B) Area to Asymptote

Ex:

$$\int_0^4 \frac{1}{\sqrt{x}}\,dx =$$

$$= \lim_{a\to 0^+} 2\sqrt{x}\Big|_a^4 = \lim_{a\to 0^+}\left(4-2\sqrt{a}\right) = 4$$

$\boxed{\text{converges to } 4}$

Ex:

$$\int_0^1 \ln x\,dx =$$

$u = \ln x \quad dv = dx$
$du = \frac{1}{x} \quad v = x$

$$\lim_{a\to 0^+} x\ln x - x\Big|_a^1 = \lim_{a\to 0^+} -1 - (a\ln a - a) =$$

$$\lim_{a\to 0^+} \frac{\ln a}{a^{-1}}$$

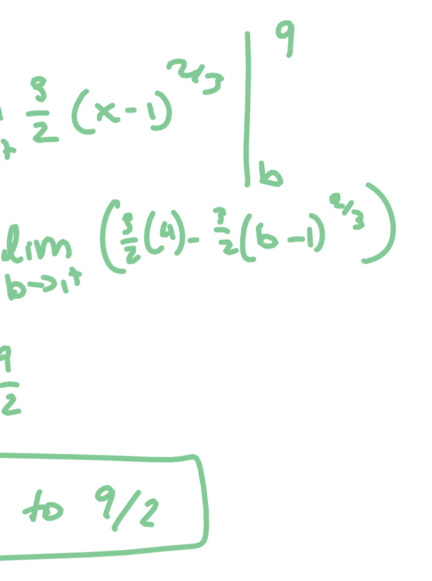
$$\overset{\frac{\infty}{\infty}}{=} LH \lim_{a\to 0^+} \frac{1}{a}\cdot -a^2 = 0$$

$= -1$

$\boxed{\text{converges to } -1}$

Ex:

$$\int_0^9 (x-1)^{-1/3}\,dx =$$



$$\int_0^1 (x-1)^{-1/3}\,dx + \int_1^9 (x-1)^{-1/3}\,dx$$

$$= \lim_{a\to 1^-} \frac{3}{2}(x-1)^{2/3}\Big|_0^a + \lim_{b\to 1^+} \frac{3}{2}(x-1)^{2/3}\Big|_b^9$$

$$= \lim_{a\to 1^-}\left(\frac{3}{2}(a-1)^{2/3} - \frac{3}{2}\right) + \lim_{b\to 1^+}\left(\frac{3}{2}(4) - \frac{3}{2}(b-1)^{2/3}\right)$$

$$-\frac{3}{2} + \frac{12}{2} = \frac{9}{2}$$

$\boxed{\text{converges to } 9/2}$

Ex:

$$\int_3^\infty \frac{1}{x\sqrt{x^2-9}}\,dx =$$

$$\lim_{a\to 3^+} \frac{1}{3}\operatorname{arcsec}\frac{|x|}{3}\Big|_a^4 + \lim_{b\to\infty} \frac{1}{3}\operatorname{arcsec}\frac{|x|}{3}\Big|_4^b$$

$$\lim_{a\to 3^+}\left(\cancel{\frac{1}{3}\operatorname{arcsec}\left(\frac{4}{3}\right)} - \frac{1}{3}\operatorname{arcsec}\left(\frac{a}{3}\right)\right)$$

$\to 0$

$+$

$$\lim_{b\to\infty}\left(\frac{1}{3}\operatorname{arcsec}\frac{b}{3} - \cancel{\frac{1}{3}\operatorname{arcsec}\left(\frac{4}{3}\right)}\right)$$

$\to \frac{1}{3}\cdot\frac{\pi}{2}$

$0 + \frac{1}{3}\cdot\frac{\pi}{2}$

$\boxed{\text{Converges at } \frac{1}{3}\cdot\frac{\pi}{2}}$